ARDHYA NOVRI RAMADHAN

IoT Developer | Embedded Systems and Monitoring

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Bachelor of Engineering in Electrical Engineering from Diponegoro University (GPA 3.43/4.00), with hands-on ESP32 sensor prototypes, STM32 data acquisition and Python monitoring interfaces. Internship experience covers IoT temperature monitoring, DC motor control and airport electrical systems. Interested in supporting IoT deployment, monitoring and troubleshooting for industrial operations.

TECHNICAL SKILLS

Embedded and sensors: C, STM32 HAL, STM32CubeIDE, ESP32, DS18B20, SEN0240, ADC, DMA, timers and UART.

Monitoring and software: Python, Tkinter, live plots, CSV/JSON data handling, digital filtering, RMS/MAV extraction, Git and automated tests.

Connectivity and IoT: Blynk and Thinger.io project experience; HTTP/JSON APIs with FastAPI; introductory MQTT concepts.

Field exposure: UPS/converters, protection-relay learning, SCADA instruction, sensor installation and operating documentation.

RELEVANT EXPERIENCE

Internship Trainee Jul - Aug 2025

PT Angkasa Pura Indonesia | Soekarno-Hatta International Airport

- Completed an eight-week rotation: five weeks in UPS/converter systems and three weeks in electrical protection.
- Visited substations and focused on protection relays. Received SCADA instruction and observed PLC installations.

Electronics Technician Jan - Feb 2025

Redesma Technologies | Semarang

- Built ESP32 and DS18B20 temperature-monitoring, DC motor-control and UART communication prototypes for an internship seminar.
- Co-authored an SOP comparing Blynk and Thinger.io for real-time temperature monitoring.

Basic Electronics Practicum Assistant 2024 - 2025

Diponegoro University

- Served for two consecutive semesters in the 2024/2025 academic year. Guided students through laboratory activities and basic electronic components.

CORE TECHNICAL PROJECT

EMG Acquisition and Monitoring System Thesis | 2026

ardhyanovri.my.id/projects/emg-thesis.html

- Implemented 12-bit ADC acquisition with DMA on STM32F411CEU6 and SEN0240 at a nominal sampling rate of 4 kHz.
- Applied 20 Hz high-pass, 500 Hz low-pass and 50 Hz notch filters; extracted RMS/MAV features over 400-sample windows.
- Built the Python/Tkinter GUI for UART telemetry at 115200 baud, live plots, device status and CSV export. Teammates handled stimulation and fuzzy control.

ADDITIONAL RELEVANT PROJECTS

Solar-Powered Smart Owl House KKN | Jan - Feb 2026

ardhyanovri.my.id/projects/kkn-solar.html

- Helped assemble and install the solar-powered system in Mutihan, Klaten, including sensors, battery and controller.
- Prepared operating and maintenance guidance for village use and handover.

uart-fixture-kit Oct 2026

github.com/vanovri/uart-fixture-kit

- Built synthetic UART fixtures and virtual-time replay for fragmented, concatenated, malformed and truncated byte streams.
- Made parser failures reproducible through a Python API, CLI and automated checks, without requiring serial hardware.

DataRescue Oct 2026

github.com/vanovri/datarescue

- Built a local Python/FastAPI and React workbench for CSV/JSON validation, explicit normalization and data-quality review.
- Preserved original values, quarantined invalid records and exported clean data with row-level audit reports.

CURRENT TECHNICAL ROLE

CTO / Technical Lead Oct 2026 - Present

ENVIID Carbon Solutions

- Sole developer of the company website, built with AI assistance, including project information, a carbon calculator and a digital MRV workflow demonstration.
- Worked with structured inputs, validation and HTTP/JSON endpoints in the calculator and workflow demonstrations.

EDUCATION

Diponegoro University: Bachelor of Engineering in Electrical Engineering | Aug 2022 - Sep 2026

GPA: 3.43/4.00. Graduated 29 September 2026.

SELECTED TRAINING

- Introduction to Internet of Things, Cisco Networking Academy. Completed 27 September 2026.
- Automation Clinic: Motor Protection and Control, Engineering Academy. September 2024.
- Introduction to Electronics, The Open University OpenLearn. Completed a 10-hour course in May 2024.
- IT: Device to Device Communication, The Open University OpenLearn. Completed a 20-hour course in May 2024.

COLLABORATION AND AVAILABILITY

Head of Media, SEED UNDIP (2025); electronics workshop instructor, ENIGMA Vol. 3 (October 2024). Experience explaining technical material and working with student and community teams.

Based in Bekasi. Available from 1 November 2026 for the East Jakarta placement and the 24-month contract indicated in the application form.

Development of software projects and the thesis GUI used AI coding assistance.